

```
(Employee._fields + Manager._fields))
>>> EmployeeManager._fields
('name', 'age', 'address', 'manager_name', 'manager_
phone')
```

deque

`deque` or a `double-ended-queue` is a generalized data structure for stacks and queues. It is quite similar to the lists in python but provides much faster and memory efficient `append` and `pop` operations which are thread safe.

A `deque` can be initialized with two parameters, one being the initial iterable we wish to add to the `deque`; if not provided, an empty deque is returned. The other parameter being the `maxlength`, which defines the bounded length of the deque. If `maxlength` is not specified, deque may grow up to any arbitrary length, but otherwise once a `deque` is full, adding any more elements at one end will discard an equal number of elements from the other end.

```
>>> dq = deque([1,2,3], 2)
>>> dq
deque([2, 3], maxlen=2)
```

Since we specified `maxlen` as 2 even though we provided three elements during initialization. It retains only 2 elements, discards the rest.

Performing operations on deque:

- Insertions on a deque can be performed from either end using `append()` or `appendleft()` method. As the name suggests, `appendleft` will add a new element to the left end of the deque while the `append` method does the same on the right end.

```
>>> dq = deque([1,2,3])
>>> dq.append(4)
>>> dq
deque([1, 2, 3, 4])
>>> dq.appendleft(5)
>>> dq
deque([5, 1, 2, 3, 4])
```

- Removal or deletion of an element from the `deque` can be done using a `pop()` and a `popleft()` operation, to remove an element from the `right` and the `left` end respectively.